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20118 Traveler Circle

Walnut, CA 91789

DOB: Feb. 25, 2003

JEREMY J.X. HSIAO

*Driven by the fusion of technology, data, and sound,
with an interest in computational systems for artistic expression.*

EDUCATION

STANFORD UNIVERSITY | BA, MA | CUM GPA: 3.667

Sept 2021 - Jun 2026

- Bachelor of Arts in Communications
- Double Minors: Computer Science, Music

EXPERIENCE

STUDENT RESEARCH | Remote + Stanford, CA

Aug 2024 - Apr 2025

Center for Computer Research in Music and Acoustics (CCRMA) under Adjunct Prof. Malcolm Slaney

- Co-authored implementation and optimization of Loudness Model for Time Varying Sounds with Binaural Inhibition, translating from MATLAB to Python with NumPy, JAX, and SciPy. Added comprehensive testing suite, demonstration scripts, and optimized processes.
- Auditory Attention Decoder: Implemented Backward TRF model to predict and evaluate listeners' attention toward attended versus unattended musical instruments with multi-channel signal processing on existing EEG data.

Communication Capstone Project under Prof. Nilam Ram

Mar 2025 – Jun 2025

- Full-Stack Application: Developed a web-app visualization platform after collecting data from 197 experimental conditions across 44 studies in EEG and music, with React.js.
- Authored an accompanying full-length research paper, organizing a foundation for future work.

COURSE PROJECTS | Stanford, CA

Sept 2022 - Present

- Data Sonification:** NASA wildfire and wind data from the 2025 LA wildfires into musical performances.
- Audiovisual Installations/Performances:** Projection Mapping, Logic Pro X, reactive MAX Patches, audio generation, live performances. Available on my portfolio: jerhsiao.github.io
- Quantum Computer Music Generation:** Analyzes short input, generates one minute of quantumly random harmonic ambient music with Qiskit, audio programming, algorithms, and visualization.
- Huffman Encoding:** Huffman Encoding Tree, Map, Decoding, Handled I/O Streams
- Memory Heap Allocator:** Implicit Free List Allocator and Explicit Free List Allocator
- MovieBot:** Built Model to Recommend Movies and Alternative MovieBot Using OpenAI's API

PERSONAL PROJECTS

Jan 2024 - Present

- N-Back Position Test Application:** A comprehensive, configurable web application for administering the N-Back position cognitive task to measure working memory performance. <https://n-back-nine.vercel.app/>
- Word Hunt Solver:** A fast and intuitive solver for a 4x4 Word Hunt game. With DFS, it quickly finds and highlights English words within the grid and displays with an intuitive UI.
- Email Bot:** Automated mass, personalized email marketing for a small e-commerce business using Google Cloud Console's Gmail API, reaching 22,000 potential clients

SKILLS

Programming: Python, Java, C, C++, CSS, R, MATLAB, MAX/ MSP, Chuck, JavaScript

Libraries and Frameworks: NumPy, SciPy, PyTorch, JAX, Librosa, Matplotlib, React, Qiskit, D3.js, XGBoost

Audio Signal Processing: Spectral Analysis, Onset Detection, Real-Time Audio Processing

Research: Experimental Design, Data Analysis, Statistical Methods, Causal Inference

Tools: GitHub, Colab, Visual Studio, Xcode, Web Development

WORK EXPERIENCE

- **SOMUNICATE PROJECT** | Technische Universität Berlin **Jun 2025 – Sept 2025**
Leading research as first author on onset detection classification methods for academic publication. Joining the Audio Communication Department at TU Berlin to develop neural network models that predict user perception of UX interaction sounds. Improved audio segmentation algorithm through evaluation of onset detection. Developed comprehensive validation frameworks using ground truth datasets and identified specific algorithmic improvements that enhanced classification accuracy.
- **STANFORD MUSIC LIBRARY STUDENT WORKER** | Stanford, CA **Jan 2022 – Jun 2025**
Managed library database and operations. Processed books, sheet music, vinyl's, CDs, headphones, returns, checkouts, arrivals, transfers, etc.
- **YUNUS & YOUTH FELLOWSHIP WRITER** | Remote **Jun 2024 – Dec 2024**
Conducted 13 interviews to write engaging, in-depth narrative articles and profiles about social entrepreneur fellows and the story of their social enterprise for the Y&Y Fellowship.
- **WARNER BROS. PICTURES MUSIC INTERN** | Burbank, CA **Jun 2022- Aug 2023**
Automated compilation of 500+ music industry contacts, significantly reducing manual processing time. Managed and updated music cue data, script breakdowns, production schedules, licensing data
- **WARNER BROS. TV PUBLICITY INTERN** | Burbank, CA **Jun 2021 - Aug 2021**
Management for WB entities' social media accounts. Increased TikTok followers on @warnerbrostv by 22%. Overall management of 100+ accounts, with a focus on 6. Produced 10-20 posts per week.

ACTIVITIES

- STANFORD CONCERT NETWORK** | Stanford, CA **Sept 2021 - Mar 2022**
 - Successfully negotiated contracts with venues and an artist's management team to organize SCN's New Member Showcase concerts with a \$2500 budget
- LIVE SOUND TECHNICIAN** | Vineyard of Hope Church | Walnut, CA **Jul 2021 - Jan 2022**
 - Live mixing of drums, keyboard, guitars, bass, and vocals
 - Set up and operation of audio equipment (mics, 32 input Yamaha live mixer, etc.)
- STANFORD SIGMA CHI ALPHA OMEGA CHAPTER** | Stanford, CA **Sept 2022 - Present**
 - Historian: Responsible for collection and protection of the Chapter history through social media, scrapbooks, organized documentation and events.

HONORS AND AWARDS

2020 Elks Scholar - Most Valuable Student Scholarship
2020 Coca-Cola Scholars Semifinalist
2019 826 National's Poets in Revolt. Poet for anthology created with Amanda Gorman and Kate Deciccio.
2019 14th Eugene O'Neill Theater Center Young Playwrights Festival Semifinalist
2018 National Student Poets Program Semifinalist (1 of 35 out of 23,000)